

Reshaping data

Warmup activity

Work on the warmup activity (on the course website) with a neighbor, then we will discuss as a class

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
  <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA    NA    NA
NA    NA
2 Albania      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
3 Algeria      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
4 Andorra      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
```

Question: What does a row in this data represent?

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
  <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA    NA    NA
NA    NA
2 Albania      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
3 Algeria      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
4 Andorra      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
```

Question: What does a row in this data represent?

Each row is one country

Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
  <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA    NA    NA
NA    NA
2 Albania      NA    NA    NA    NA    NA    NA    NA    NA
NA    NA
3 Algeria      NA    NA    NA    NA    NA    NA    NA    NA
NA    NA
4 Andorra      NA    NA    NA    NA    NA    NA    NA    NA
NA    NA
```

Can I use the data in this form to make a scatterplot or fit a linear model?

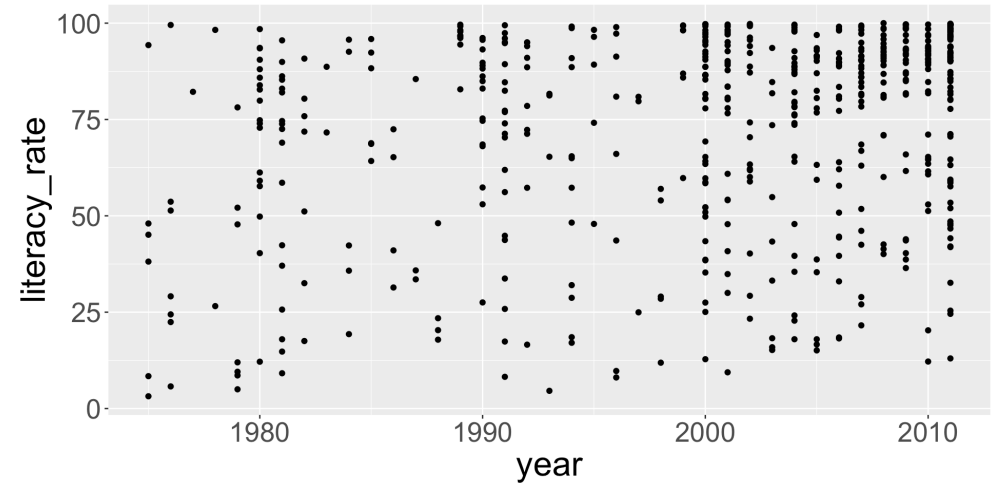
Warmup

```
# A tibble: 260 × 38
  country `1975` `1976` `1977` `1978` `1979` `1980` `1981` `1982`
`1983` `1984`
  <chr>    <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
<dbl> <dbl>
1 Afghan...    NA    NA    NA    NA    4.99    NA    NA    NA
NA    NA
2 Albania      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
3 Algeria      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
4 Andorra      NA    NA    NA    NA    NA      NA    NA    NA
NA    NA
```

What do I want my data to look like instead?

What we want

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	...	...	...



```
1 litF_long |>
2   ggplot(aes(x = year,
3               y = literacy_rate)) +
4   geom_point()
```

What we want

	country	year	literacy_rate
1	Afghanistan	1979	4.987460
2	Afghanistan	2011	13.000000
3	Albania	2001	98.252274
4	Albania	2008	94.681814
5	Albania	2011	95.691480
6	Algeria	1987	35.839915
7	Algeria	2002	60.075082
8	Algeria	2006	63.918785
9	Angola	2001	54.194488
10	Angola	2011	58.608460
11	Anguilla	1984	95.714930
12	Antigua and Barbuda	2001	99.420000
13	Antigua and Barbuda	2011	99.420000
14	Argentina	1980	93.580894
15	Argentina	1991	96.041358
16	Argentina	2001	97.193411
17	.	.	.

(Intercept)	year
-1323.2098674	0.6979597

```
1 lm(literacy_rate ~ year, data = litF_long)
```


Tidy data

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174604898
China	1999	212258	1272915272
China	2000	216766	1280425583

variables

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174604898
China	1999	212258	1272915272
China	2000	216766	1280425583

observations

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20595360
Brazil	1999	37737	172006362
Brazil	2000	80488	174604898
China	1999	212258	1272915272
China	2000	216766	1280425583

values

(Image from *R for Data Science*)

Tidy data

Literacy data in tidy form:

```
# A tibble: 571 × 3
```

	country	year	literacy_rate
	<chr>	<chr>	<dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8
7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

Another example

Data on three patients (A, B, C), with two blood pressure measurements (bp1 and bp2) per patient:

	id	bp1	bp2
1	A	100	120
2	B	140	115
3	C	120	125

How might we want to reshape this data?

Another example

Original data:

	id	bp1	bp2
1	A	100	120
2	B	140	115
3	C	120	125

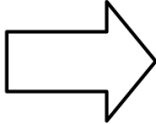
Reshaped data:

	id	measurement	value
1	A	bp1	100
2	A	bp2	120
3	B	bp1	140
4	B	bp2	115
5	C	bp1	120
6	C	bp2	125

Question: how do we do this reshaping in R?

Reshaping data: `pivot_longer`

id	bp1	bp2
A	100	120
B	140	115
C	120	125



id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

```
1 df |>
2   pivot_longer(
3     cols = bp1:bp2,
4     names_to = "measurement",
5     values_to = "value"
6   )
```

(Image and example from *R for Data Science*)

pivot_longer

A tibble: 260 × 38

	country	`1975`	`1976`	`1977`	`1978`	`1979`	`1980`	`1981`	`1982`	`1983`	`1984`
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Afghan...	NA	NA	NA	NA	4.99	NA	NA	NA	NA	NA
2	Albania	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
3	Algeria	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
4	Andorra	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

```
1 litF |>
2   pivot_longer(
3     cols = ...,
4     names_to = ...,
5     values_to = ...
6   )
```

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate"
6   )
```

A tibble: 9,620 × 3

	country	year	literacy_rate
	<chr>	<chr>	<dbl>
1	Afghanistan	1975	NA
2	Afghanistan	1976	NA
3	Afghanistan	1977	NA
4	Afghanistan	1978	NA
5	Afghanistan	1979	4.99
6	Afghanistan	1980	NA
7	Afghanistan	1981	NA
8	Afghanistan	1982	NA
9	Afghanistan	1983	NA
10	Afghanistan	1984	NA

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate"
6   ) |>
7   drop_na()
```

A tibble: 571 × 3

	country <chr>	year <chr>	literacy_rate <dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8
7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

pivot_longer

```
1 litF |>
2   pivot_longer(
3     cols = -country,
4     names_to = "year",
5     values_to = "literacy_rate",
6     values_drop_na = T
7   )
```

A tibble: 571 × 3

	country	year	literacy_rate
	<chr>	<chr>	<dbl>
1	Afghanistan	1979	4.99
2	Afghanistan	2011	13
3	Albania	2001	98.3
4	Albania	2008	94.7
5	Albania	2011	95.7
6	Algeria	1987	35.8
7	Algeria	2002	60.1
8	Algeria	2006	63.9
9	Angola	2001	54.2
10	Angola	2011	58.6

Example 2

Now consider the following table:

```
1 ex_df
```

	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

What will the following code return?

```
1 ex_df |>
2   pivot_longer(cols = -id,
3                 names_to = "group_obs",
4                 values_to = "value")
```

Example 2

Original data:

```
1 ex_df
```

	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

Reshaped data:

```
1 ex_df |>
2   pivot_longer(cols = -id,
3                 names_to = "group_obs",
4                 values_to = "value")
```

A tibble: 12 × 3

	id	group_obs	value
	<dbl>	<chr>	<dbl>
1	1	x_1	3
2	1	x_2	5
3	1	y_1	0
4	1	y_2	2
5	2	x_1	1
6	2	x_2	8
7	2	y_1	1
8	2	y_2	7
9	3	x_1	4
10	3	x_2	9
11	3	y_1	2
12	3	y_2	9

Example 2

Original data:

```
1 ex_df
```

	id	x_1	x_2	y_1	y_2
1	1	3	5	0	2
2	2	1	8	1	7
3	3	4	9	2	9

Reshaped data:

```
1 ex_df |>
2   pivot_longer(cols = -id,
3                 names_to = c("group", "obs"),
4                 names_sep = "_",
5                 values_to = "value")
```

A tibble: 12 × 4

	id	group	obs	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	x	1	3
2	1	x	2	5
3	1	y	1	0
4	1	y	2	2
5	2	x	1	1
6	2	x	2	8
7	2	y	1	1
8	2	y	2	7
9	3	x	1	4
10	3	x	2	9
11	3	y	1	2
12	3	y	2	9

Example 3

Consider the following example data:

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

What if we want the data to look like this:

A tibble: 12 × 4

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115
7	2	hr	1	75
8	2	hr	2	81
9	3	bp	1	125

Example 3

```
1 df_3
```

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

```
1 df_3 |>  
2   pivot_longer(cols = -id,  
3                 names_to = c("measurement", "stage"),  
4                 names_sep = "_",  
5                 values_to = "value")
```

```
# A tibble: 12 × 4
```

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115

7	2 hr	1	75
8	2 hr	2	81
9	3 bp	1	125
10	2 hr	2	120

Example 3

```
1 df_3 |>
2   pivot_longer(cols = -id,
3                 names_to = c("measurement", "stage"),
4                 names_sep = "_",
5                 values_to = "value")
```

Step 1: Pivot

A tibble: 6 × 3

	id	measurement	value
	<dbl>	<chr>	<dbl>
1	1	bp_1	100
2	1	bp_2	120
3	1	hr_1	60
4	1	hr_2	77
5	2	bp_1	120
6	2	bp_2	115

Step 2: Separate columns

A tibble: 6 × 4

	id	measurement	stage	value
	<dbl>	<chr>	<chr>	<dbl>
1	1	bp	1	100
2	1	bp	2	120
3	1	hr	1	60
4	1	hr	2	77
5	2	bp	1	120
6	2	bp	2	115

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 5 in *R for Data Science* (2nd ed.)